I. SUPPLEMENTARY MATERIAL: DERIVATION OF TIME INDEPENDENT VELOCITY POTENTIAL FOR A TWO-LAYER SYSTEM IN A RECTANGULAR DOMAIN

To derive the velocity potential (ϕ) for each medium i, we need to solve the Laplace equation which goes by:

$$\Delta\widehat{\phi}_i = \frac{\partial^2\widehat{\phi}_i}{\partial x^2} + \frac{\partial^2\widehat{\phi}_i}{\partial y^2} + \frac{\partial^2\widehat{\phi}_i}{\partial z^2}. \tag{1}$$

Using separation of variables, we get

$$\widehat{\phi}_i(x, y, z) = X_i(x)Y_i(y)Z_i(z), \tag{2}$$

leading to

$$\frac{1}{X_i}\frac{\mathrm{d}^2X_i}{\mathrm{d}x^2} + \frac{1}{Y_i}\frac{\mathrm{d}^2Y_i}{\mathrm{d}y^2} = -\frac{1}{Z_i}\frac{\mathrm{d}^2Z_i}{\mathrm{d}z^2} = -k_z^2, \tag{3}$$

where $k_z^2 = k_x^2 + k_y^2$. We write the function $Z_i(z)$ as

$$Z_i(z) = Z_i^1 e^{k_z} + Z_i^2 e^{-k_z}. \tag{4}$$

The boundary conditions pertaining to the z-plane include,

$$\left.\frac{\partial\phi_1}{\partial z}\right|_{z=h_1} = \left.\frac{\partial\phi_2}{\partial z}\right|_{z=-h_2} = 0, \tag{5}$$

$$\begin{aligned} \therefore Z_1(z) &= Z_1\cosh\left[k_z(z-h_1)\right], \\ Z_2(z) &= Z_2\cosh\left[k_z(z+h_2)\right]. \end{aligned} \tag{6}$$

Normalizing the above equation to get,

$$\begin{aligned} Z_1 &= \frac{\omega}{k_z}\frac{\cosh[k_z(z-h_1)]}{\sinh[k_zh_1]}, \\ Z_2 &= -\frac{\omega}{k_z}\frac{\cosh[k_z(z+h_2)]}{\sinh[k_zh_2]}. \end{aligned} \tag{7}$$

The boundary conditions with respect to the x and y plane are,

$$\begin{aligned} \left.\frac{\partial X_1}{\partial x}\right|_{x=\frac{L_x}{2}} &= \left.\frac{\partial X_2}{\partial x}\right|_{x=-\frac{L_x}{2}} = 0, \\ \left.\frac{\partial Y_1}{\partial y}\right|_{y=\frac{L_y}{2}} &= \left.\frac{\partial Y_2}{\partial y}\right|_{y=-\frac{L_y}{2}} = 0. \end{aligned} \tag{8}$$

Concerning the present coordinate system with the origin present at the center of the quadrilateral system, cosine functions are introduced so that the conditions at the frontal and

the sagittal surfaces are met. To make the potential zero in planes $x = (\pm L_x/2)$ and $y = (\pm L_y/2)$, it is necessary that

$$\begin{aligned} X_1 = X_2 = \cos\left[k_x\left(x + \frac{L_x}{2}\right)\right], \quad k_x = \frac{m\pi}{L_x}, \\ Y_1 = Y_2 = \cos\left[k_y\left(y + \frac{L_y}{2}\right)\right], \quad k_y = \frac{n\pi}{L_y}. \end{aligned} \tag{9}$$

Inserting Eq. (7) and Eq. (9) in Eq. (2), we get the following equation for the time independent velocity potential in a two layer stratified system:

$$\begin{bmatrix} \widehat{\phi}_1 \\ \\ \widehat{\phi}_2 \end{bmatrix} = A_{mn} \begin{bmatrix} \dfrac{\omega}{k}\dfrac{\cosh[k(z-h_1)]}{\sinh[kh_1]} \\ -\dfrac{\omega}{k}\dfrac{\cosh[k(z+h_2)]}{\sinh[kh_2]} \end{bmatrix} \cos\left[\frac{m\pi}{L_x}\left(x + \frac{L_x}{2}\right)\right] \cos\left[\frac{n\pi}{L_y}\left(y + \frac{L_y}{2}\right)\right], \tag{10}$$

where $k = k_z$, the mode dependent wave number. The the rate of change of interface elevation is dependent on the normal derivative of the velocity potential such that,

$$\left.\frac{\partial \phi_1}{\partial z}\right|_{z=0} = \left.\frac{\partial \phi_2}{\partial z}\right|_{z=0} = \left.\frac{\partial \eta}{\partial t}\right|_{z=0}. \tag{11}$$

Therefore, the time independent interface elevation $\widehat{\eta}$ is

$$\widehat{\eta} = \mathrm{i}A_{mn}\cos\left[\frac{m\pi}{L_x}\left(x + \frac{L_x}{2}\right)\right] \cos\left[\frac{n\pi}{L_y}\left(y + \frac{L_y}{2}\right)\right]. \tag{12}$$

II. SUPPLEMENTARY MATERIAL: DERIVATION OF TIME INDEPENDENT ELECTRIC POTENTIAL FOR A TWO-LAYER SYSTEM IN A RECTANGULAR DOMAIN

The Laplace equation for the electric potential reads:

$$\Delta\widehat{\varphi} = \frac{\partial^2\widehat{\varphi}}{\partial x^2} + \frac{\partial^2\widehat{\varphi}}{\partial y^2} + \frac{\partial^2\widehat{\varphi}}{\partial z^2}. \tag{13}$$

We write the separation ansatz,

$$\widehat{\varphi}_i(x, y, z) = X_i(x)Y_i(y)Z_i(z)H, \tag{14}$$

where, H is the dependence of the electrical potential on the elevation of the interface η. Similar to the steps involved for the velocity potential in Sec. I, we solve for $Z_i(z)$, $X_i(x)$ and $Y_i(y)$ as follows:

$$\underbrace{\frac{1}{X_i}\frac{\mathrm{d}^2X_i}{\mathrm{d}x^2}}_{-k_x^2} + \underbrace{\frac{1}{Y_i}\frac{\mathrm{d}^2Y_i}{\mathrm{d}y^2}}_{-k_y^2} + \underbrace{\frac{1}{Z_i}\frac{\mathrm{d}^2Z_i}{\mathrm{d}z^2}}_{k_z^2} = 0, \tag{15}$$

Applying the boundary conditions pertaining to the top and bottom surfaces are

$$\begin{aligned} \widehat{\varphi}_1 &= 0|_{z=h_1}, \\ J_{2,z} &= 0|_{z=-h_2} = 0, \end{aligned} \tag{16}$$

we get the solution to $Z_i(z)$ to be

$$\begin{aligned} Z_1(z) &= -Z_1 \sinh[k_z(z-h_1)], \\ Z_2(z) &= Z_2 \cosh[k_z(z+h_2)]. \end{aligned} \tag{17}$$

After normalization:

$$\begin{aligned} Z_1(z) &= -\frac{\sinh[k_z(z-h_1)]}{\sigma_1 \cosh[k_z h_1]} \\ Z_2(z) &= -\frac{\cosh[k_z(z+h_2)]}{\sigma_2 \sinh[k_z h_2]} \end{aligned} \tag{18}$$

Similarly, $X_i(x)$ and $Y_i(y)$ amount to

$$\begin{aligned} X_1 = X_2 &= \cos\left[k_x\left(x+\frac{L_x}{2}\right)\right], \quad k_x = \frac{m\pi}{L_x}, \\ Y_1 = Y_2 &= \cos\left[k_y\left(y+\frac{L_y}{2}\right)\right], \quad k_y = \frac{n\pi}{L_y}. \end{aligned} \tag{19}$$

Inserting Eq. (17) and Eq. (19) in Eq. (14), we get the formulation

$$\begin{aligned} \widehat{\varphi}_1 &= -A_{mn}\frac{\sinh[k(z-h_1)]}{\sigma_1 \cosh[kh_1]} \cos\left[\frac{m\pi}{L_x}\left(x+\frac{L_x}{2}\right)\right] \cos\left[\frac{n\pi}{L_y}\left(y+\frac{L_y}{2}\right)\right] H, \\ \widehat{\varphi}_2 &= -A_{mn}\frac{\cosh[k(z+h_2)]}{\sigma_2 \sinh[kh_2]} \cos\left[\frac{m\pi}{L_x}\left(x+\frac{L_x}{2}\right)\right] \cos\left[\frac{n\pi}{L_y}\left(y+\frac{L_y}{2}\right)\right] H. \end{aligned} \tag{20}$$

Here, $k = k_z$. The jump condition at the interface reads

$$J_0(\sigma_2^{-1} - \sigma_1^{-1})\widehat{\eta} = \widehat{\varphi}_2|_{z=0} - \widehat{\varphi}_1|_{z=0}. \tag{21}$$

Solving for H by substituting Eq. (20) in Eq. (21) gives

$$H = \frac{\mathrm{i}J_0(\sigma_1^{-1} - \sigma_2^{-1})}{\sigma_1^{-1}\tanh[kh_1] + \sigma_2^{-1}\coth[kh_2]} \tag{22}$$

The equation of the time independent electric potential for a two-layer stratified system reads

$$\begin{bmatrix} \widehat{\varphi}_1 \\ \\ \widehat{\varphi}_2 \end{bmatrix} = \frac{A_{mn}J_0(\sigma_1^{-1} - \sigma_2^{-1})\widehat{\eta}}{\sigma_1^{-1}\tanh[kh_1] + \sigma_2^{-1}\coth[kh_2]} \begin{bmatrix} -\dfrac{\sinh[k(z-h_1)]}{\sigma_1 \cosh[kh_1]} \\ -\dfrac{\cosh[k(z+h_2)]}{\sigma_2 \sinh[kh_2]} \end{bmatrix} \tag{23}$$

III. SUPPLEMENTARY MATERIAL: FORMULATION OF THE FREDHOLM ALTERNATIVE

To recap, the two-wave perturbation ansatz reads

$$[\boldsymbol{u}_i, p_i, \eta, \boldsymbol{j}_i] = \left(C\left[\widehat{\boldsymbol{u}}_i, \widehat{p}_i, \widehat{\eta}, \widehat{\boldsymbol{j}}_i\right] + C'\left[\widehat{\boldsymbol{u}}_i', \widehat{p}_i', \widehat{\eta}', \widehat{\boldsymbol{j}}_i'\right] + \left[\widetilde{\boldsymbol{u}}_i, \widetilde{p}_i, \widetilde{\eta}, \widetilde{\boldsymbol{j}}_i\right]\right) \mathrm{e}^{\left(\alpha + \mathrm{i}\bar{\omega}\right)t}. \tag{24}$$

The balance equations considering only leading order terms, for a given wave number (m, n) are

$$\rho_i(\mathrm{i}\omega)\widehat{\boldsymbol{u}}_i + \boldsymbol{\nabla}\widehat{p}_i = 0, \tag{25a}$$

$$\boldsymbol{\nabla} \cdot \widehat{u}_i = 0, \tag{25b}$$

$$\widehat{\boldsymbol{j}}_i = -\sigma_i \boldsymbol{\nabla}\widehat{\varphi}_i, \qquad \boldsymbol{\nabla} \cdot \widehat{\boldsymbol{j}}_i = 0. \tag{25c}$$

The complete expression when we input the ansatz [Eq. (24)] in the momentum balance equation is given as

$$\begin{aligned} &C\big(\mathrm{i}\bar{\omega} + \alpha\big)\rho\widehat{\boldsymbol{u}} + C'\big(\mathrm{i}\bar{\omega} + \alpha\big)\rho\widehat{\boldsymbol{u}}' + (\mathrm{i}\bar{\omega} + \alpha)\widetilde{\boldsymbol{u}} + \boldsymbol{\nabla}\widetilde{p} + (\mathrm{i}\bar{\omega} + \alpha)\widetilde{\boldsymbol{u}}' + \boldsymbol{\nabla}\widetilde{p}' \\ &= C\Big(\widehat{\boldsymbol{j}} \times B_z\boldsymbol{e}_z\Big) + C'\Big(\widehat{\boldsymbol{j}}' \times B_z\boldsymbol{e}_z\Big) \end{aligned} \tag{26}$$

On expanding $\bar{\omega}$ on the LHS of Eq. (26), we get the following:

$$\begin{aligned} \Longrightarrow\ & C\big(\mathrm{i}\omega - \mathrm{i}\delta\omega + \alpha\big)\rho\widehat{\boldsymbol{u}} + C'\big(\mathrm{i}\omega - \mathrm{i}\delta\omega + \alpha\big)\rho\widehat{\boldsymbol{u}}' \\ & + (\mathrm{i}\omega - \mathrm{i}\delta\omega + \alpha)\widetilde{\boldsymbol{u}} + \boldsymbol{\nabla}\widetilde{p} + (\mathrm{i}\omega - \mathrm{i}\delta\omega + \alpha)\widetilde{\boldsymbol{u}}' + \boldsymbol{\nabla}\widetilde{p}', \end{aligned} \tag{27}$$

We now make a certain leading order approximation $\omega\widetilde{\boldsymbol{u}}_i \gg \delta\omega\widetilde{\boldsymbol{u}}_i$, to simplify Eq. (27) as

$$\begin{aligned} &C\big(-[\mathrm{i}\delta\omega] + \alpha\big)\rho\widehat{\boldsymbol{u}} + C'\big([\mathrm{i}\delta\omega] + \alpha\big)\rho\widehat{\boldsymbol{u}}' + (\mathrm{i}\omega)\widetilde{\boldsymbol{u}} + \boldsymbol{\nabla}\widetilde{p} + (\mathrm{i}\omega')\widetilde{\boldsymbol{u}}' + \boldsymbol{\nabla}\widetilde{p}' \\ &= C\Big(\widehat{\boldsymbol{j}} \times B_z\boldsymbol{e}_z\Big) + C'\Big(\widehat{\boldsymbol{j}}' \times B_z\boldsymbol{e}_z\Big), \end{aligned} \tag{28}$$

$$\begin{aligned} &\Longrightarrow (\mathrm{i}\omega)\rho\widetilde{\boldsymbol{u}} + \boldsymbol{\nabla}\widetilde{p} + (\mathrm{i}\omega')\rho\widetilde{\boldsymbol{u}}' + \boldsymbol{\nabla}\widetilde{p}' \\ &= C\Big(\widehat{\boldsymbol{j}} \times B_z\boldsymbol{e}_z + \Big([\mathrm{i}\delta\omega] - \alpha\Big)\rho\widehat{\boldsymbol{u}}\Big) + C'\Big(\widehat{\boldsymbol{j}}' \times B_z\boldsymbol{e}_z + \Big(-[\mathrm{i}\delta\omega] - \alpha\Big)\rho\widehat{\boldsymbol{u}}'\Big). \end{aligned} \tag{29}$$

Applying the Fredholm alternative to the equations over the two fluid regions as follows:

$$\Sigma_{i=1,2} \int_{V_i} \left[\widehat{\boldsymbol{u}}_i^* \cdot [\text{Eq. (29)}] + \widehat{p}_i^*(\boldsymbol{\nabla} \cdot \widetilde{\boldsymbol{u}}_i)\right] \mathrm{d}V, \tag{30}$$

we can write the inner product (for quantities pertaining to wave number (m,n)) as

$$\begin{aligned}&\textstyle\sum_{i=1,2}\int_{V_i}\Big[C\widehat{\boldsymbol{u}}_i^*\cdot\Big([\mathrm{i}\omega]\rho\widetilde{\boldsymbol{u}}_i+\boldsymbol{\nabla}\widetilde{p}_i\Big)+C\widehat{p}_i^*(\boldsymbol{\nabla}\cdot\widetilde{\boldsymbol{u}}_i)\Big],\\&=\textstyle\sum_{i=1,2}\int_{V_i}C\widehat{\boldsymbol{u}}_i^*\cdot\Big[C\Big(\widehat{\boldsymbol{j}}_i\times B_z\boldsymbol{e}_z+\big([\mathrm{i}\delta\omega]-\alpha\big)\rho\widehat{\boldsymbol{u}}_i\Big)\Big].\end{aligned}\tag{31}$$

The left-hand side of the Eq. (31) can be expanded to

$$\int_{V_i}C\widehat{\boldsymbol{u}}_i^*(\boldsymbol{\nabla}\widetilde{p}_i)=-C\int_{V_i}(\boldsymbol{\nabla}\cdot\widehat{\boldsymbol{u}}_i^*)\,\widehat{p}_i+C\oint_S\widehat{\boldsymbol{u}}_i^*\widetilde{p}_i\cdot\boldsymbol{n}\,\mathrm{d}S,\tag{32}$$

and

$$\int_{V_i}C\widehat{p}_i^*(\boldsymbol{\nabla}\cdot\widetilde{\boldsymbol{u}}_i)=C\int_{V_i}-\boldsymbol{\nabla}\widehat{p}_i^*\widetilde{\boldsymbol{u}}_i+C\oint_S\widehat{p}_i^*\widetilde{\boldsymbol{u}}_i\cdot\boldsymbol{n}\,\mathrm{d}S,\tag{33}$$

where $\widetilde{\boldsymbol{u}}\cdot\boldsymbol{n}$ can be written as $\widetilde{\boldsymbol{u}}_z|_{z=0}$. Also we can write

$$\partial_t\eta=\widetilde{\boldsymbol{u}}_z|_{z=0}\Longrightarrow\mathrm{i}\omega_{11}\widetilde{\eta}+C\big(\alpha\widehat{\eta}-[\mathrm{i}\delta\omega]\widehat{\eta}\,\big).\tag{34}$$

Therefore, from the leading order balance Eqs. (25a) and (25b), we deduce

$$\begin{aligned}&\textstyle\int_{V_i}C\Big[\widetilde{\boldsymbol{u}}_i\cancelto{0}{\big([\mathrm{i}\omega]\rho\widehat{\boldsymbol{u}}_i^*-\boldsymbol{\nabla}\widehat{p}_i^*\big)}-\cancelto{0}{(\boldsymbol{\nabla}\cdot\widehat{\boldsymbol{u}}_i^*)}\,\widehat{p}_i\Big]+C\oint_S(\widehat{p}_i^*\widetilde{\boldsymbol{u}}_i+\widetilde{p}_i\widehat{\boldsymbol{u}}_i^*)\cdot\boldsymbol{n}\,\mathrm{d}S,\\&\Longrightarrow\big(\alpha-[\mathrm{i}\delta\omega]\big)C^2\underbrace{\oint_S(\widehat{p}_2^*-\widehat{p}_1^*)|_{z=0}\,\widehat{\eta}\,\mathrm{d}S}_{K_{\eta\eta}}.\end{aligned}\tag{35}$$

The dynamic boundary condition for gravity capillary waves is given by:

$$-\gamma_{\mathrm{int}}\nabla^2\eta+(\rho_2-\rho_1)g\eta=p_2\,\big|_{z=0}-p_1\,\big|_{z=0}.\tag{36}$$

Incorporating the above equation in the surface integral of the pressure jump gives

$$\textstyle\oint_S(\widehat{p}_2^*-\widehat{p}_1^*)|_{z=0}\widehat{\eta}\,\mathrm{d}S\;=\;\int_{\mathcal{S}}\Big(-\gamma_{\mathrm{int}}|\boldsymbol{\nabla}\widehat{\eta}|^2+(\rho_2-\rho_1)g|\widehat{\eta}|^2\Big)\Big|_{z=0}\mathrm{d}S=K_{\eta\eta}.\tag{37}$$

By substituting Eq. (37) in (35), we obtain

$$\begin{aligned}&\forall\,(m,n),\;\;C\textstyle\oint_S(\widehat{p}_i^*\widetilde{\boldsymbol{u}}_i+\widetilde{p}_i\widehat{\boldsymbol{u}}_i^*)\cdot\boldsymbol{n}\,\mathrm{d}S\to\big(\alpha-[\mathrm{i}\delta\omega]\big)C^2K_{\eta\eta},\\&\forall\,(m',n'),\;\;C'\textstyle\oint_S(\widehat{p}_i'^*\widetilde{\boldsymbol{u}}_i'+\widetilde{p}_i'\widehat{\boldsymbol{u}}_i'^*)\cdot\boldsymbol{n}\,\mathrm{d}S\to\big(\alpha+[\mathrm{i}\delta\omega]\big)C'^2K_{\eta'\eta'}.\end{aligned}\tag{38}$$

On the right-hand side of the Eq. (31), we can write the non-Lorentz force terms as

$$\int_{V_i}C\widehat{\boldsymbol{u}}_i^*\cdot\Big[C\big([\mathrm{i}\delta\omega]-\alpha\big)\rho\widehat{\boldsymbol{u}}_i\Big]=C^2\big([\mathrm{i}\delta\omega]-\alpha\big)\int_{V_i}\rho_i|\widehat{\boldsymbol{u}}_i|^2\mathrm{d}V.\tag{39}$$

Integration by parts on Eq. (39) considering the potential wave solution of $\widehat{p} = -\rho_i(\mathrm{i}\omega)\widehat{\phi}_i$ gives

$$\begin{aligned}
\sum_{i=1,2}\int_{V_i}\rho_i|\widehat{\boldsymbol{u}}_i|^2\mathrm{d}V &= \rho_1\oint_{\delta V_1}\widehat{\phi}_1^*\boldsymbol{\nabla}\widehat{\phi}_1\cdot\boldsymbol{n}_1\mathrm{d}S + \rho_2\oint_{\delta V_2}\widehat{\phi}_2^*\boldsymbol{\nabla}\widehat{\phi}_2\cdot\boldsymbol{n}_2\mathrm{d}S,\\
&= \oint_{\mathcal{S}}\Big(-\mathrm{i}\rho_1\overline{\omega}\widehat{\phi}_1^* + \mathrm{i}\rho_2\overline{\omega}\widehat{\phi}_2^*\Big)\Big|_{z=0}\widehat{\eta}\mathrm{d}S,\\
&= \oint_{\mathcal{S}}\big(\widehat{p}_2^* - \widehat{p}_1^*\big)\Big|_{z=0}\widehat{\eta}\mathrm{d}S \Longrightarrow K_{\eta\eta}.
\end{aligned} \tag{40}$$

Solving Eqs. (38) and (39) gives the matrix equations given below:

$$\big(\alpha - [\mathrm{i}\delta\omega]\big)C^2K_{\eta\eta} = C^2I_{\eta\eta} + CC'I_{\eta\eta'} + C^2\big([\mathrm{i}\delta\omega] - \alpha\big)K_{\eta\eta}, \tag{41}$$

$$\big(\alpha + [\mathrm{i}\delta\omega]\big)C'^2K_{\eta'\eta'} = CC'I_{\eta'\eta} + C'^2I_{\eta'\eta'} + C'^2\big(-[\mathrm{i}\delta\omega] - \alpha\big)K_{\eta'\eta'}. \tag{42}$$

The above two equations form the matrix to solve for α.

IV. SUPPLEMENTARY MATERIAL: SOLVING THE MECHANICAL ENERGY BALANCE EQUATION FOR VISCOUS DECAY

The energy balance equation is as follows:

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(E_c + E_p\big) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\sum_{i=1,2}\frac{1}{2}\int_{V_i}\rho_i|\boldsymbol{u}_i|^2\mathrm{d}V]\right) + \frac{\mathrm{d}}{\mathrm{d}t}\left(\int_S\left[\frac{1}{2}(\rho_2-\rho_1)g\eta^2 + \gamma\sqrt{1+|\nabla\eta|^2}\right]\mathrm{d}S,\right. \tag{43}$$

$$\mathcal{D} = \sum_{i=1,2} 2\rho_i\nu_i\int_{V_i}\boldsymbol{\varepsilon}_i : \boldsymbol{\varepsilon}_i\,\mathrm{d}V. \tag{44}$$

We write the velocity and the elevation profile of a decaying rotating wave given in Eq. (43), ignoring the surface tension terms as

$$\left.\begin{aligned} \boldsymbol{u}_i &\approx [\widehat{\boldsymbol{u}}_i\,\mathrm{e}^{\mathrm{i}\omega t} + \widehat{\boldsymbol{u}}_i\,\mathrm{e}^{-\mathrm{i}\omega t}]\mathrm{e}^{-\lambda t} = \widehat{\boldsymbol{u}}_i\,\cos[\omega]\,\mathrm{e}^{-\lambda t} \\ \eta &\approx [\widehat{\eta}\,\mathrm{e}^{\mathrm{i}\omega t} - \widehat{\eta}\,\mathrm{e}^{-\mathrm{i}\omega t}]\mathrm{e}^{-\lambda t} = \widehat{\eta}\,\sin[\omega]\,\mathrm{e}^{-\lambda t} \end{aligned}\right\}. \tag{45}$$

In a standing wave, the total energy always equals the sum of kinetic and potential energy at a phase difference of $\pi/2$. From Eqs. (37) and (40), we write

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\big(E_{\mathrm{kin}} + E_{\mathrm{pot}}\big) &= -2\lambda\left(\frac{1}{2}K\,\cos^2[\omega]\,\mathrm{e}^{-2\lambda t} + \frac{1}{2}K\,\sin^2[\omega]\,\mathrm{e}^{-2\lambda t}\right), \\ &= -\lambda\,K\,\mathrm{e}^{-2\lambda t}, \qquad \big\{K = K_{\eta\eta} \in (m,n) \vee K_{\eta'\eta'} \in (m',n'). \end{aligned} \tag{46}$$

In Eq. (44), the strain tensor is given by $\boldsymbol{\varepsilon}_i = \frac{1}{2}\Big(\boldsymbol{\nabla}\boldsymbol{u}_i + \boldsymbol{\nabla}\boldsymbol{u}_i^{\mathrm{T}}\Big)$. Introducing the Lamb's dissipation function as

$$\mathcal{D} = \sum_{i=1,2} 2\rho_i\nu_i\int_{V_i}\boldsymbol{\varepsilon}_i : \boldsymbol{\varepsilon}_i\,\mathrm{d}V \Longrightarrow \sum_{i=1,2}\int_{V_i}\rho_i\Phi_{\mathcal{D}}\mathrm{d}V, \tag{47}$$

$$\phi_{\mathcal{D}} = \nu_i\left[\frac{1}{2}\Big(\nabla\boldsymbol{u}_i^{\mathrm{T}} - \nabla\boldsymbol{u}_i\Big)^2\right] + 2\nu_i\nabla\boldsymbol{u}_i\nabla\boldsymbol{u}_i^{\mathrm{T}}. \tag{48}$$

On simplifying $\phi_{\mathcal{D}}$, we obtain the dissipation integral as follows

$$\mathcal{D} = \sum_{i=1,2}\rho_i\nu_i\left(\underbrace{\int_{V_i}\big(\nabla\times\boldsymbol{u}_i\big)^2\mathrm{d}V}_{\mathrm{I}} - \underbrace{\int_{\delta V_i}2\big[\boldsymbol{u}_i\times(\nabla\times\boldsymbol{u}_i)\big].\boldsymbol{n}\mathrm{d}S}_{\mathrm{II}} + \underbrace{\int_{\delta V_i}\nabla|\boldsymbol{u}_i|^2\cdot\boldsymbol{n}\mathrm{d}S}_{\mathrm{III}}\right). \tag{49}$$

Integral I emphasizes the energy dissipation due to rotational flows and contributes largely to the dissipation in the system. The second integral is of higher order and considers both the bulk and rotational flows. Integral III is related to the bulk irrotational stresses (relevant

in free surface waves) and takes into account only the flow potential. We solve integral I in Eq. (49) taking velocity relations from Eqs. (**??**), (**??**) and (**??**) to obtain $\mathcal{D}_{\text{wall}}$ and $\mathcal{D}_{\text{int}}$ and integral II and III together yields $\mathcal{D}_{\text{irr}}$. The solutions for the respective dissipation terms are as follows:

$$\begin{aligned}\mathcal{D}_{\text{wall}} = \sum_{i=1,2} \Bigg[& \xi_{m\times n} \left(\frac{\rho_i \omega^2 \sqrt{\omega \nu_i}}{\sqrt{2}\, k^2} \right) \left(\frac{L_x L_y\, k^2}{2 \sinh^2[k h_i]} \right) \\ & + \xi_n \left(\frac{\rho_i \omega^2 \sqrt{\omega \nu_i}}{\sqrt{2}\, k^2} \right) \left(\frac{h_i (n^2\pi^2 - k^2 L_y^2)}{L_y\, \sinh^2[k h_i]} + \frac{(n^2\pi^2 + k^2 L_y^2)}{L_y\, k\, \tanh[k h_i]} \right) \\ & + \xi_m \left(\frac{\rho_i \omega^2 \sqrt{\omega \nu_i}}{\sqrt{2}\, k^2} \right) \left(\frac{h_i (m^2\pi^2 - k^2 L_x^2)}{L_x\, \sinh^2[k h_i]} + \frac{(m^2\pi^2 + k^2 L_x^2)}{L_x\, k\, \tanh[k h_i]} \right) \Bigg], \end{aligned} \tag{50}$$

$$\mathcal{D}_{\text{int}} = \xi_{m\times n} \frac{\omega_{11}^2 \sqrt{\omega_{11}}}{2\sqrt{2}} \frac{\left(\coth[K_{\eta\eta} h_1] + \coth[K_{\eta\eta} h_2] \right)^2}{\dfrac{1}{L_x L_y} \left(\dfrac{1}{\rho_1 \sqrt{\nu_1}} + \dfrac{1}{\rho_2 \sqrt{\nu_2}} \right)}, \tag{51}$$

$$\begin{aligned}\mathcal{D}_{\text{ext}} = & \frac{2\xi_{m\times n} (\rho_2 - \rho_1) g k^2 \left(\dfrac{1}{\rho_1 \sqrt{\nu_1}} + \dfrac{1}{\rho_2 \sqrt{\nu_2}} \right)^{-1}}{\rho_1 \coth[k h_1] + \rho_2 \coth[k h_2]} \Bigg(\Bigg[\frac{\rho_2 \sqrt{\nu_1 \nu_2} + \rho_1 \nu_1}{\rho_2 \sqrt{\nu_2} \tanh[k h_1]} + \frac{\rho_1 \sqrt{\nu_1 \nu_2} + \rho_2 \nu_2}{\rho_1 \sqrt{\nu_1} \tanh[k h_2]} \\ & - \left(\coth[k h_1] + \coth[k h_2] \right) \left(\sqrt{\nu_1} + \sqrt{\nu_2} \right) \Bigg] \Bigg), \end{aligned} \tag{52}$$

Here, $\xi_r \begin{cases} 1/4, & r > 0 \\ 1/2, & r = 0 \end{cases}.$

$$\therefore -\lambda K \mathrm{e}^{-2\lambda t} = -\mathcal{D}\, \mathrm{e}^{-2\lambda t} \Longrightarrow \lambda = \frac{\mathcal{D}}{K}. \tag{53}$$

V. DETERMINING THE CRITICAL WAVE NUMBER AND THE STABILITY ONSET

A. Choosing the lateral wave numbers

In our formulation it is not directly possible to explicitly obtain the most unstable wave number which may either refer to the onset of instability (β_{crit}) or the largest possible growth rate (max α_{MPR}). We begin with choosing arbitrary, initial wave numbers and

calculate the growth rate α_{MPR} and the critical instability onset β_{crit}, and iteratively find the wave numbers which satisfy the former condition. Though it might seem that there are infinite permutations and combinations of numbers to choose from, there are certain conditions that reduce the sample space. They are:

1. The mode selection function $\Theta > 0$ only if $m + n = \mathbb{N}_{\text{odd}}$, $m' + n' = \mathbb{N}_{\text{odd}}$

2. Large scale wave numbers considerably increase damping and the instability threshold. Therefore wave numbers up to $(m, n), (m', n') \leq 10$ can be taken into consideration

B. Calculating the stability onset

1. Calculate wave number k, k'.

$$k = \pi\sqrt{\frac{m^2}{L_x^2} + \frac{n^2}{L_y^2}}, \qquad k' = \pi\sqrt{\frac{m'^2}{L_x^2} + \frac{n'^2}{L_y^2}}.$$

2. Calculate angular frequencies ω, ω'.

$$\omega = \pm\sqrt{\frac{(\rho_2 - \rho_1)gk + \gamma_{\text{int}}k^3}{\rho_1 \coth[k\, h_1] + \rho_2 \coth[k\, h_2]}}, \qquad \omega' = \pm\sqrt{\frac{(\rho_2 - \rho_1)gk' + \gamma_{\text{int}}k'^3}{\rho_1 \coth[k'\, h_1] + \rho_2 \coth[k'\, h_2]}}.$$

3. Note: $\bar{\omega} = \dfrac{\omega + \omega'}{2}$, $\delta\omega = \dfrac{\omega - \omega'}{2}$ and $\delta_r = \begin{cases} 1, & r = 0 \\ 2, & r > 0 \end{cases}$.

4. Determine the average viscous damping, $\bar{\lambda} = \dfrac{\left(\lambda_{\text{wall}} + \lambda_{\text{int}} + \lambda_{\text{irr}}\right) + \left(\lambda'_{\text{wall}} + \lambda'_{\text{int}} + \lambda'_{\text{ext}}\right)}{2}$,

$$\lambda_{\text{wall}} = \sum_{i=1,2} \frac{\dfrac{1}{\sqrt{2}}\dfrac{\rho_i\sqrt{\omega\nu_i}}{kL_xL_y}}{\left[\dfrac{\rho_1}{\tanh[kh_1]} + \dfrac{\rho_2}{\tanh[kh_2]}\right]} \left(\left[\frac{L_xL_y\, k^2}{2\sinh^2[kh_i]} + \frac{\delta_m h_i(n^2\pi^2 - k^2L_y^2)}{2L_y\, \sinh^2[kh_i]}\right.\right.$$
$$\left.\left. + \frac{\delta_m(n^2\pi^2 + k^2L_y^2)}{2L_y\, k\, \tanh[kh_i]} + \frac{\delta_n h_i(m^2\pi^2 - k^2L_x^2)}{2L_x\, \sinh^2[kh_i]} + \frac{\delta_n(m^2\pi^2 + k^2L_x^2)}{2L_x\, k\, \tanh[kh_i]}\right]\right),$$

$$\lambda_{\text{int}} = \frac{k\sqrt{\omega}}{2\sqrt{2}} \frac{\left(\coth[kh_1] + \coth[kh_2]\right)^2}{\left(\frac{1}{\rho_1\sqrt{\nu_1}} + \frac{1}{\rho_2\sqrt{\nu_2}}\right)\left[\frac{\rho_1}{\tanh[kh_1]} + \frac{\rho_2}{\tanh[kh_2]}\right]},$$

$$\lambda_{\text{irr}} = \frac{2k^2\left(\frac{1}{\rho_1\sqrt{\nu_1}} + \frac{1}{\rho_2\sqrt{\nu_2}}\right)^{-1}}{\rho_1\coth\left[kh_1\right] + \rho_2\coth\left[kh_2\right]}\left(\left[\frac{\rho_2\sqrt{\nu_1\nu_2} + \rho_1\nu_1}{\rho_2\sqrt{\nu_2}\tanh\left[kh_1\right]} + \frac{\rho_1\sqrt{\nu_1\nu_2} + \rho_2\nu_2}{\rho_1\sqrt{\nu_1}\tanh\left[kh_2\right]}\right.\right.$$
$$\left.\left. - \left(\coth\left[kh_1\right] + \coth\left[kh_2\right]\right)\left(\sqrt{\nu_1} + \sqrt{\nu_2}\right)\right]\right).$$

5. The onset of instability occurs now can be calculated by

$$\beta_{\text{crit}} = \frac{L_x L_y}{h_1 h_2 \Theta}\sqrt{\frac{\bar{\lambda}^2 - \left[\mathrm{i}\delta\omega - \delta\lambda\right]^2}{(\bar{\omega}^2 - \delta\omega^2)\mathcal{T}\mathcal{T}'}}.$$

6. The growth rate

$$\alpha = \pm\sqrt{J^2 {B_z}^2 \Theta^2\,(\bar{\omega}^2 - \delta\omega^2)\,\frac{\mathcal{T}\mathcal{T}'}{\mathcal{U}\mathcal{U}'} + \left[\mathrm{i}\delta\omega - \delta\lambda\right]^2} \;-\; \bar{\lambda},$$

where

$$\Theta = \pm\sqrt{\delta_{m\times m'}\,\delta_{n\times n'}}\,\frac{(-1^{m+m'} - 1)(-1^{n+n'} - 1)(m^2 n'^2 - m'^2 n^2)}{(m^2 - m'^2)(n^2 - n'^2)},$$

$$\mathcal{T} = \frac{\Lambda}{k(k^2 - k'^2)L_x L_y}\left[\frac{k}{\tanh[k'h_2]} - \frac{k'}{\tanh[kh_2]} + \frac{k'\,\operatorname{sech}[k'h_1]}{\sinh[kh_1]} - \frac{k'}{\tanh[kh_1]} + k\tanh[k'h_1]\right],$$

$$\mathcal{T}' = \frac{\Lambda'}{k'(k'^2 - k^2)L_x L_y}\left[\frac{k'}{\tanh[kh_2]} - \frac{k}{\tanh[k'h_2]} + \frac{k\,\operatorname{sech}[kh_1]}{\sinh[k'h_1]} - \frac{k}{\tanh[k'h_1]} + k'\tanh[kh_1]\right],$$

$$\mathcal{U} = \left(\gamma_{\text{int}}\,k^2 + (\rho_2 - \rho_1)g\right), \qquad \mathcal{U}' = \left(\gamma_{\text{int}}\,k'^2 + (\rho_2 - \rho_1)g\right),$$

$$\Lambda = \frac{(\sigma_1^{-1} - \sigma_2^{-1})}{\sigma_1^{-1}\,\tanh[kh_1] + \sigma_2^{-1}\,\coth[kh_2]}, \qquad \Lambda' = \frac{(\sigma_1^{-1} - \sigma_2^{-1})}{\sigma_1^{-1}\,\tanh[k'h_1] + \sigma_2^{-1}\,\coth[k'h_2]}.$$

7. Repeat step (1) to obtain the lowest possible β_{crit} or the highest possible α for all wave number combinations.